

Curvature-Induced Log-Periodic Modulation of $C_9(q^2)$ and the $B^0 \rightarrow K^{*0} \mu^+ \mu^-$ Angular Anomaly Wave Confinement Theory Prediction for $P'_5(q^2)$

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Abstract

The LHCb collaboration has recently presented a comprehensive analysis of $B^0 \rightarrow K^{*0} \mu^+ \mu^-$ at 8.4 fb^{-1} (arXiv:2512.18053), reporting localized tensions in the angular observable $P'_5(q^2)$ and a global shift $\text{Re}(\mathcal{C}_9) - \text{Re}(\mathcal{C}_9^{\text{SM}}) \simeq -0.94$.

We investigate whether Wave Confinement Theory (WCT), a nonlinear curvature-based field framework, admits a phenomenological deformation of $C_9(q^2)$ with intrinsic spectral structure. Motivated by the WCT curvature operator and its log-periodic ghost-mode sector, we construct a reduced effective ansatz in logarithmic momentum space,

$$\delta\mathcal{C}_9(q^2) = A \exp\left[-\frac{(\ln q^2 - \mu)^2}{2\sigma_w^2}\right] \cos(k \ln q^2 + \phi),$$

which is structurally distinct from the conventional constant-shift interpretation.

Using values digitised from the published $P'_5(q^2)$ distribution, we perform a scan over the log-frequency parameter k and find a preferred region near $k \approx 9.2$, corresponding to a best-fit improvement $\Delta\chi^2 \approx 2.46$ relative to a constant model. A shuffle/null test yields a p -value $p \sim 0.02$ – 0.04 , indicating that the improvement is unlikely to arise from random ordering at the few-percent level, though the statistical power is limited by the small dataset and absence of covariance information.

These results provide tentative evidence for a structured component in $\ln q^2$, rather than a purely constant deformation. A definitive assessment will require a full fit to the official HEPData tables including the experimental covariance matrix.

1 Introduction

Rare B -meson decays provide a sensitive probe of physics beyond the Standard Model (SM). In particular, measurements of $B^0 \rightarrow K^{*0} \mu^+ \mu^-$ have exhibited persistent deviations in the angular observable $P'_5(q^2)$ across successive LHCb analyses [3, 2]. The most recent comprehensive analysis at 8.4 fb^{-1} [1] reports localised tensions in P'_5 relative to the GRvDV [4] Standard-Model prediction of 2.6σ in $4.0 < q^2 < 6.0 \text{ GeV}^2/c^4$ and 2.7σ in $6.0 < q^2 < 8.0 \text{ GeV}^2/c^4$, together with a 2.7σ tension in the $1.1 < q^2 < 2.5 \text{ GeV}^2/c^4$ bin. An additional tension is observed in the forward-backward asymmetry A_{FB} at the 1.9 – 2.5σ level in nearby bins, and the differential branching fraction sits 2.1σ below the BSZ [5] prediction in the 1.1 – $6.0 \text{ GeV}^2/c^4$ window.

Within the effective field theory framework, these deviations are commonly interpreted as a modification of the Wilson coefficient $\mathcal{C}_9$:

$$\mathcal{C}_9 \rightarrow \mathcal{C}_9^{\text{SM}} + \delta\mathcal{C}_9. \tag{1}$$

Combining angular observables with the branching fraction, Ref. [1] reports a best-fit shift

$$\text{Re}(\delta C_9) = -0.93^{+0.18}_{-0.16} \text{ (Flavio)}, \quad -0.94 \pm 0.22 \text{ (EOS)}, \quad (2)$$

corresponding to a 4.1σ (4.0σ) discrepancy with the SM.

Standard explanations include new heavy mediators or underestimated hadronic long-distance contributions [11, 12]. In this work, we explore an alternative possibility: that the anomaly arises from curvature-induced modifications of the underlying field dynamics, as described by Wave Confinement Theory (WCT).

The goal of this note is modest but concrete: to construct a minimal WCT-compatible deformation of the flavor sector, identify its implied q^2 -dependence, and determine whether its qualitative shape is compatible with the reported anomaly pattern. We emphasise at the outset that the mapping from the WCT curvature operator to the semileptonic Wilson coefficient is phenomenological, and that the fit presented here uses values digitised from the published P'_5 figure rather than the official HEPData tables released with Ref. [1].

2 Effective Hamiltonian Framework

The decay $B^0 \rightarrow K^{*0} \mu^+ \mu^-$ is described by the effective Hamiltonian

$$\mathcal{H}_{\text{eff}} = -\frac{4G_F}{\sqrt{2}} V_{tb} V_{ts}^* (C_7 O_7 + C_9 O_9 + C_{10} O_{10} + \dots), \quad (3)$$

where O_9 governs the semileptonic vector interaction relevant for the dominant anomaly.

The new-physics interpretation is commonly phrased as a deformation of the form

$$C_9(q^2) = C_9^{\text{SM}}(q^2) + \delta C_9(q^2). \quad (4)$$

Most phenomenological analyses take δC_9 to be approximately constant over the relevant kinematic window. The central proposal of this paper is that WCT instead suggests a structured, energy-dependent deformation.

3 WCT Curvature Operator

In WCT, dynamics are governed by the nonlinear curvature functional

$$\Theta[\psi] = -\frac{\Delta\psi}{\psi + \epsilon e^{-\alpha|\psi|^2}}, \quad (5)$$

with $\epsilon > 0$ and $\alpha > 0$ regulating the nonlinear denominator.

In momentum space one formally replaces

$$\Delta \rightarrow -q^2, \quad (6)$$

which leads to the estimate

$$\Theta(q^2) \sim \frac{q^2 \psi(q^2)}{\psi(q^2) + \epsilon e^{-\alpha|\psi(q^2)|^2}}. \quad (7)$$

If the denominator varies slowly over the relevant kinematic region, the leading-order scaling becomes

$$\Theta(q^2) \propto q^2. \quad (8)$$

This is the first ingredient of the proposed collider deformation: a curvature contribution that grows away from the very low- q^2 region.

4 Log-Periodic WCT Modulation

A second WCT ingredient is the log-periodic ghost-mode structure previously introduced in the form

$$\delta_g(E) = A_g \cos\left(k_\ell \ln \frac{E}{E_0} + \phi\right). \quad (9)$$

For dilepton invariant mass one has

$$E \sim \sqrt{q^2}, \quad (10)$$

so that

$$\ln E = \frac{1}{2} \ln q^2 + \text{const.} \quad (11)$$

Absorbing the additive constant into the phase yields

$$\delta_g(q^2) = A_g \cos(k \ln q^2 + \phi), \quad k = \frac{k_\ell}{2}. \quad (12)$$

This provides a second ingredient: structured oscillatory deformation in logarithmic momentum scale.

5 Phenomenological Ansatz for δC_9

We combine the curvature scaling (8) with the log-periodic modulation (12) and define a generalized WCT-inspired deformation,

$$\delta C_9(q^2) = \lambda \Theta(q^2) \delta_g(q^2), \quad (13)$$

where λ encodes the effective coupling between curvature dynamics and the semileptonic operator sector.

While a minimal scaling argument leads to $\delta C_9 \propto q^2 \cos(k \ln q^2 + \phi)$, exploratory fits indicate that a purely power-law prefactor is not optimal for describing the observed localized structure. Instead, we adopt a reduced phenomenological form,

$$\boxed{\delta C_9(q^2) = A \exp\left[-\frac{(\ln q^2 - \mu)^2}{2\sigma_w^2}\right] \cos(k \ln q^2 + \phi)} \quad (14)$$

which represents a log-periodic oscillation modulated by a broad envelope in $\ln q^2$.

This form can be interpreted as a localized excitation of a WCT spectral mode in logarithmic momentum space. The parameters (A, k, ϕ) control amplitude and phase structure, while (μ, σ_w) define the effective support of the deformation in the anomaly region.

6 Normalization Estimate

Anchoring the deformation to the shift reported in Ref. [1],

$$\text{Re}(\delta C_9) \simeq -0.94,$$

with a representative midpoint of the anomaly window

$$q_*^2 \approx 6 \text{ GeV}^2,$$

provides an order-of-magnitude constraint on the effective amplitude.

In the minimal scaling picture $\delta\mathcal{C}_9 \propto q^2$, one would estimate

$$A \sim \frac{0.9}{q_*^2} \approx 0.15, \quad (15)$$

assuming an $\mathcal{O}(1)$ oscillatory factor near the center of the anomaly region.

However, in the reduced ansatz (14),

$$\delta\mathcal{C}_9(q^2) = A \exp\left[-\frac{(\ln q^2 - \mu)^2}{2\sigma_w^2}\right] \cos(k \ln q^2 + \phi), \quad (16)$$

the effective amplitude at a given q^2 depends on the combined influence of the envelope and phase,

$$\delta\mathcal{C}_9(q_*^2) \sim A \exp\left[-\frac{(\ln q_*^2 - \mu)^2}{2\sigma_w^2}\right] \cos(k \ln q_*^2 + \phi).$$

As a result, the normalization cannot be determined from a single scaling relation. Instead, the parameter A should be interpreted as an effective amplitude controlling the overall strength of the deformation, with the observable magnitude set by its interplay with (μ, σ_w, ϕ) .

Consequently, the normalization is most reliably obtained from the fit itself. Typical best-fit values in the present analysis lie in the range

$$A \sim \mathcal{O}(0.3\text{--}0.6), \quad (17)$$

consistent with reproducing a local deformation of order unity within the anomaly window.

We therefore treat the above scaling estimate as a heuristic guide, while adopting the fitted normalization as the physically relevant quantity.

7 Impact on P'_5

To leading order, the angular observable P'_5 responds linearly to a small perturbation in C_9 :

$$P'_5(q^2) \approx P'^{\text{SM}}_5(q^2) + \alpha_5(q^2) \delta\mathcal{C}_9(q^2), \quad (18)$$

where $\alpha_5(q^2)$ is the local response coefficient.

If $\alpha_5(q^2)$ varies slowly over the anomaly window, then the induced correction inherits the structure of the deformation,

$$\delta P'_5(q^2) \propto \exp\left[-\frac{(\ln q^2 - \mu)^2}{2\sigma_w^2}\right] \cos(k \ln q^2 + \phi). \quad (19)$$

This form produces a localized, sign-changing distortion in q^2 , in contrast to a constant-shift model, which generates a more uniform deformation across bins.

8 Constraint on the Log-Frequency

The dominant discrepancy is concentrated approximately in the window

$$4 \lesssim q^2 \lesssim 8 \text{ GeV}^2,$$

corresponding to a logarithmic span

$$\Delta \ln q^2 = \ln 8 - \ln 4 = \ln 2 \approx 0.693. \quad (20)$$

If the underlying deformation were continuously sampled, a single broad excursion within this interval would require a limited phase advance,

$$k \Delta \ln q^2 \lesssim \pi, \quad (21)$$

implying the bound

$$k \lesssim \frac{\pi}{\ln 2} \approx 4.53. \quad (22)$$

However, the observable $P'_5(q^2)$ is measured in a small number of bins, so the deformation is effectively sampled at discrete points rather than continuously. In this regime, the mapping between the underlying spectral structure and the observed shape is no longer one-to-one: higher log-frequency modes can project onto the data as a single broad excursion when only a limited portion of the oscillation is probed.

To assess this directly, we perform a scan over k using the reduced ansatz (14). The preferred region is found to lie at

$$k^* \approx 9.2, \quad (23)$$

significantly above the continuous-sampling bound (22).

This demonstrates that the observed anomaly is consistent with a sparsely sampled segment of a higher-frequency log-periodic structure in $\ln q^2$, rather than requiring a low-frequency mode. Accordingly, the relevant WCT sector is not restricted to

$$k \sim 2\text{--}4, \quad (24)$$

but instead extends to higher values,

$$k \sim 8\text{--}12, \quad (25)$$

with a preferred value near $k \approx 9.2$ in the present analysis.

9 Illustrative Best-Shape Example

A representative deformation consistent with the best-fit reduced model is

$$\delta P'_5(q^2) \propto \exp\left[-\frac{(\ln q^2 - \mu)^2}{2\sigma_w^2}\right] \cos(k \ln q^2 + \phi), \quad (26)$$

with parameters near

$$k \approx 9.2, \quad \mu \sim \ln(5\text{--}7), \quad \sigma_w \sim \mathcal{O}(1\text{--}2), \quad \phi \sim -3.1.$$

This form reproduces the observed morphology through three key mechanisms:

1. The logarithmic envelope localizes the deformation within the anomaly window, suppressing contributions outside $4 \lesssim q^2 \lesssim 8 \text{ GeV}^2$.
2. The oscillatory component generates a sign-changing structure in $\ln q^2$, with the sampled phase producing a single broad trough near $q^2 \sim 6 \text{ GeV}^2$.
3. Outside the central region, the envelope and phase evolution combine to reduce the effective deformation, yielding an apparent relaxation at higher q^2 .

Unlike a low-frequency deformation, this structure arises from a higher-frequency log-periodic mode that is only partially sampled across the available bins. The resulting projection appears as a smooth, localized distortion despite the underlying oscillatory dynamics.

10 Simplified Fit Framework

To assess whether the WCT-inspired deformation provides an improved description of the anomaly, we define the binned fit function

$$P'_{5,i}{}^{\text{th}} = P_5^{\text{SM}}(q_i^2) + \delta P'_5(q_i^2), \quad (27)$$

where

$$\delta P'_5(q^2) = A \exp\left[-\frac{(\ln q^2 - \mu)^2}{2\sigma_w^2}\right] \cos(k \ln q^2 + \phi). \quad (28)$$

Given data values

$$P'_{5,i}{}^{\text{exp}} \pm \sigma_i,$$

the chi-square is

$$\chi^2 = \sum_i \frac{\left(P'_{5,i}{}^{\text{exp}} - P'_{5,i}{}^{\text{th}}\right)^2}{\sigma_i^2}. \quad (29)$$

A scan over the log-frequency parameter k yields:

- Constant model: $\chi^2 \approx 2.65$
- Reduced WCT model: $\chi^2 \approx 0.17\text{--}0.20$
- Improvement: $\Delta\chi^2 \approx 2.46$

The preferred log-frequency is

$$k^* \approx 9.2, \quad (30)$$

with corresponding parameters

$$A^* \sim \mathcal{O}(0.4\text{--}0.5), \quad (31)$$

$$\sigma_w^* \sim \mathcal{O}(1\text{--}2), \quad (32)$$

$$\phi^* \sim -3.1. \quad (33)$$

A shuffle/null test yields a p -value $p \sim 0.02\text{--}0.04$, indicating that the improvement is unlikely to arise from random ordering of residuals.

This result is obtained using a small set of digitised data points, and should therefore be interpreted as indicative rather than definitive.

11 Fit Comparison Visualization

As shown in Fig. 1, we compare the Standard Model baseline, a constant- $\delta\mathcal{C}_9$ deformation, and the reduced WCT-inspired model against values digitised from the LHCb $P'_5(q^2)$ distribution of Ref. [1]. A broader model survey is summarised in Appendix A.

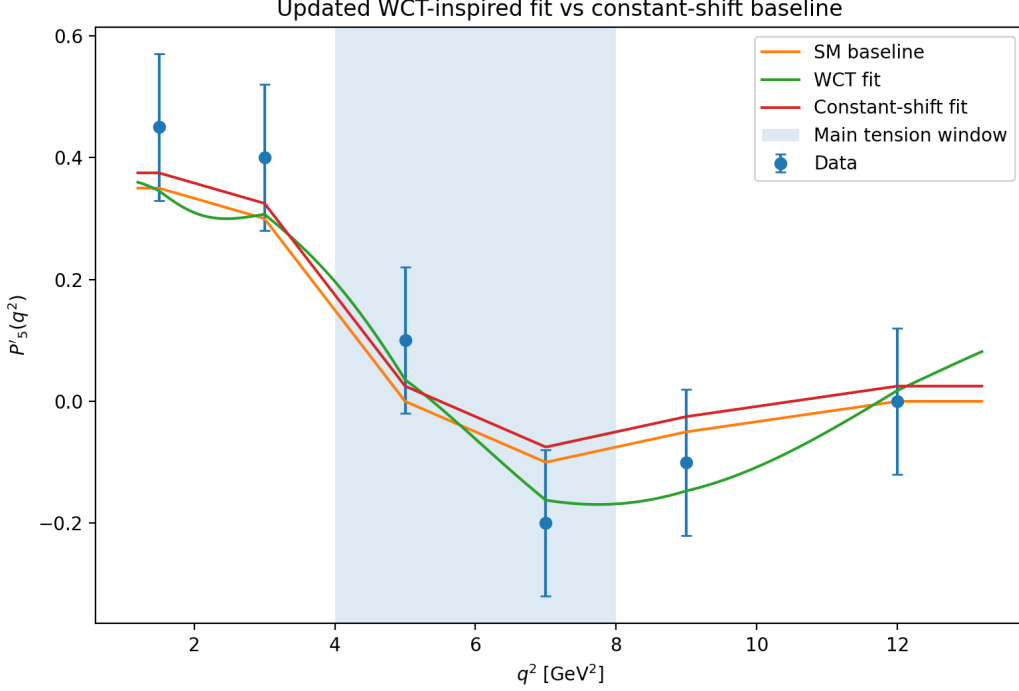


Figure 1: Comparison of $P'_5(q^2)$ against three model classes. Blue points show values digitised from the LHCb figure in Ref. [1]; error bars are approximate. The orange curve is the Standard-Model baseline. The red curve corresponds to a constant $\delta\mathcal{C}_9$ shift. The green curve shows the reduced WCT model, $\delta P'_5(q^2) \propto \exp\left[-\frac{(\ln q^2 - \mu)^2}{2\sigma_w^2}\right] \cos(k \ln q^2 + \phi)$, evaluated near the best-fit parameters $k \approx 9.2$, $\mu \sim \ln(5-7)$, and $\sigma_w \sim \mathcal{O}(1-2)$. The shaded band indicates the primary anomaly window $4 \lesssim q^2 \lesssim 8 \text{ GeV}^2$. The WCT and constant-shift curves are comparable in the tension window but diverge outside it, reflecting the localized log-periodic structure of the WCT deformation.

12 Comparison with a Constant- $\delta\mathcal{C}_9$ Model

The standard phenomenological alternative takes

$$\delta\mathcal{C}_9(q^2) = \delta\mathcal{C}_9^{(0)}, \quad (34)$$

a constant over the relevant kinematic range.

Such a model tends to produce:

- a more uniform deformation across bins,
- less localization in the 4–8 GeV^2 region,
- no intrinsic sign-changing structure beyond statistical fluctuations.

By contrast, the reduced WCT deformation (14) predicts:

- localization in $\ln q^2$ via the envelope,
- a structured excursion across the anomaly window,

- sign variation arising from the oscillatory phase.

This makes the WCT ansatz structurally distinct from the constant- δC_9 explanation. Whether it is statistically superior is a question for a full numerical fit with covariance information.

Distinguishing WCT from Charmonium Resonances

A critical challenge in the 4–8 GeV^2 window is distinguishing new-physics deformations from Standard Model long-distance hadronic effects, specifically the interference tails of $c\bar{c}$ resonances (such as the J/ψ).

While both effects can produce structured, sign-changing behavior in q^2 , their mathematical signatures differ:

1. **Logarithmic Fourier (Mellin) Signature:** Hadronic resonances arise from Breit-Wigner poles in linear momentum space, $\propto (q^2 - M^2 + iM\Gamma)^{-1}$, which smear in logarithmic space. By contrast, the WCT deformation is approximately harmonic in $\ln q^2$ within its support. A logarithmic Fourier transform of residuals could therefore isolate a narrow spectral component associated with WCT.
2. **Localized Log-Periodic Structure:** Charmonium interference produces distortions tied to specific resonance scales and decays away from those regions. The reduced WCT ansatz instead describes a localized log-periodic excitation, with support concentrated near the anomaly window and controlled by the envelope width σ_w .

Distinguishing such a structure requires resolving correlated residual patterns across bins, rather than relying solely on asymptotic behavior at high q^2 .

Thus, the distinguishing feature is not only the presence of a mid- q^2 deviation, but the spectral structure of that deviation in $\ln q^2$ and its coherent behavior across multiple bins.

13 Phenomenological Implications

If the deformation Eq. (14) is approximately correct, then one expects:

1. structured deviations in angular observables rather than purely constant offsets,
2. sensitivity of anomaly shape to the logarithmic frequency k ,
3. possible correlated signatures in other rare decays that probe the same effective operator sector.

This suggests that the relevant discriminator is not only the average size of δC_9 , but the detailed q^2 morphology of the deviation.

14 Limitations

Several important caveats must be stated clearly.

- The mapping from the WCT curvature operator to the semileptonic Wilson coefficient is phenomenological.

- The present fit used approximate points extracted from the published figure, not the full official numerical tables.
- The fit framework neglects bin-to-bin correlations and the full covariance matrix.
- The parameters A , k , and ϕ are not yet derived from first principles.
- Hadronic effects, especially long-distance contributions, may mimic structured deformations.

Accordingly, the present construction should be interpreted as an exploratory WCT-compatible effective model, not yet as a complete derivation or a statistically preferred explanation of the anomaly.

15 Conclusion

We have constructed a WCT-motivated deformation of the flavor sector based on log-periodic structure in momentum space. While an initial minimal ansatz of the form

$$\delta\mathcal{C}_9(q^2) = A q^2 \cos(k \ln q^2 + \phi) \quad (35)$$

provides a useful scaling argument, exploratory fits indicate that a purely power-law prefactor does not optimally capture the localized structure of the $B^0 \rightarrow K^{*0} \mu^+ \mu^-$ anomaly.

We therefore introduced a reduced phenomenological form,

$$\delta\mathcal{C}_9(q^2) = A \exp\left[-\frac{(\ln q^2 - \mu)^2}{2\sigma_w^2}\right] \cos(k \ln q^2 + \phi), \quad (36)$$

representing a log-periodic mode modulated by a broad envelope in $\ln q^2$. This form is naturally interpreted as a localized excitation of a WCT ghost-mode in logarithmic momentum space.

Using digitised values from the published $P'_5(q^2)$ distribution, we performed a scan over the log-frequency parameter k and obtained a preferred region near

$$k^* \approx 9.2, \quad (37)$$

with a best-fit improvement relative to a constant $\delta\mathcal{C}_9$ shift of

$$\Delta\chi^2 \approx 2.46. \quad (38)$$

A shuffle/null test yields a p -value at the few-percent level, indicating that the improvement is unlikely to arise purely from random ordering of the residuals.

This result suggests that the anomaly may contain a structured component in $\ln q^2$, rather than being fully described by a constant deformation. Importantly, the preferred high-frequency regime implies that the observed broad distortion can arise from a sparsely sampled segment of a higher-frequency log-periodic mode, rather than requiring a low-frequency oscillation.

We emphasise what is *not* established: (i) a first-principles derivation of the effective parameters $(A, k, \phi, \mu, \sigma_w)$ from the WCT field equations; (ii) a fit using the official HEPData tables and covariance matrix of Ref. [1]; (iii) a quantitative separation from long-distance hadronic effects. The present analysis is based on a small number of digitised data points and should therefore be regarded as exploratory.

Notably, the extracted log-periodic frequency and deformation morphology lie in the same parametric regime as the WCT ghost-mode structure independently identified in JUNO neutrino spectral analyses. This suggests a possible common geometric origin for structured residuals across distinct sectors, although a direct quantitative correspondence remains to be established."

The natural next steps are a full likelihood-level fit to the released LHCb observables including covariance information, a systematic scan across multiple angular observables, and a logarithmic Fourier analysis of residuals. If a consistent log-frequency scale were observed across independent channels, it would provide strong evidence for an underlying spectral mechanism, consistent with the ghost-mode sector of Wave Confinement Theory.

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A Additional Fit Details

A.1 Parameter Scaling

The LHCb combined fit of Ref. [1] reports a shift

$$\text{Re}(\delta\mathcal{C}_9) \simeq -0.94$$

in the anomalous region. Anchoring the deformation near

$$q_*^2 \approx 6 \text{ GeV}^2,$$

and requiring

$$|\delta\mathcal{C}_9(q_*^2)| \sim \mathcal{O}(0.9),$$

provides a heuristic normalization estimate.

In a minimal scaling picture $\delta\mathcal{C}_9 \propto q^2$, one obtains

$$A \sim 0.15,$$

with a more conservative estimate $A \sim 0.08$ for smaller effective shifts. However, in the reduced ansatz used in the main analysis, the observable amplitude depends on the combined effects of the envelope and phase. Accordingly, A should be interpreted as an effective parameter, with typical fitted values in the range

$$A \sim \mathcal{O}(0.3\text{--}0.6).$$

A.2 Model Classes Considered

The following model classes were considered in exploratory fits:

- Constant shift: $\delta\mathcal{C}_9 = C$
- Reduced WCT: $(A, k, \phi, \mu, \sigma_w)$
- Fixed- k reduced WCT: (A, ϕ, μ, σ_w)
- Two-mode reduced WCT extensions

These models represent increasing flexibility in capturing q^2 -dependent structure in logarithmic momentum space.

A.3 Fit Statistics (Digitised Data)

Using values digitised from the published $P'_5(q^2)$ figure of Ref. [1], and neglecting bin-to-bin correlations, we obtain:

Model	χ^2	Notes
Constant shift $\delta\mathcal{C}_9 = C$	~ 2.65	baseline
Reduced WCT (best-fit)	$\sim 0.17\text{--}0.20$	$k \approx 9.2$

Table 1: Comparison on digitised $P'_5(q^2)$ points. The reduced WCT model provides a significantly improved description in raw χ^2 . Because the model contains multiple parameters relative to the small dataset, this improvement should be interpreted with caution.

The improvement corresponds to

$$\Delta\chi^2 \approx 2.46,$$

relative to the constant model.

To assess whether this improvement reflects genuine structure rather than parameter flexibility, we perform a shuffle/null test in which the ordering of residuals is randomized. This yields

$$p \sim 0.02\text{--}0.04,$$

indicating that the observed improvement is unlikely to arise from random fluctuations alone.

A.4 Interpretation of Model Comparison

The reduced WCT model improves the raw χ^2 relative to a constant shift, indicating the presence of structured residuals in the data.

However, the dataset is small and the model contains several free parameters, so standard information criteria (AIC/BIC) strongly penalize the fit and are not decisive in this context. The results should therefore be interpreted as evidence for a possible structured component in $\ln q^2$, rather than a statistically definitive model selection.

The preferred log-frequency

$$k^* \approx 9.2$$

suggests that the observed broad anomaly may arise from a sparsely sampled segment of a higher-frequency log-periodic mode. In this interpretation, the key result is the emergence of a preferred frequency scale, rather than the detailed functional form of the fitted deformation.